

```
import os
os.environ["OPENAI_API_KEY"] = "sk-proj-aOrFc5Kv6j6RomRGR-ACpT3B1bkFJ_oZ1CAo2shNwf4dwJK1rVFosrrfAyrH7TL3Gpf5DnKrKA"
```

```
!pip install langchain chromadb faiss-cpu openai tiktoken langchain_openai langchain-community wikipedia
```

MMR - Maximum Marginal Relevance

```
from langchain_core.documents import Document
# Sample documents
docs = [
    Document(page_content="LangChain makes it easy to work with LLMs."),
    Document(page_content="LangChain is used to build LLM based applications."),
    Document(page_content="Chroma is used to store and search document embeddings."),
    Document(page_content="Embeddings are vector representations of text."),
    Document(page_content="MMR helps you get diverse results when doing similarity search."),
    Document(page_content="LangChain supports Chroma, FAISS, Pinecone, and more."),
]
```

```
!pip install langchain_community
```

```
Requirement already satisfied: langsmith<1.0.0,>=0.1.125 in /usr/local/lib/python3.12/dist-packages (from langchain_community)
Requirement already satisfied: numpy>=1.26.2 in /usr/local/lib/python3.12/dist-packages (from langchain_community) (2.0.2)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1 in /usr/local/lib/python3.12/dist-packages (from langchain_community)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain_community) (6.0.2)
Collecting requests<3.0.0,>=2.32.5 (from langchain_community)
  Downloading requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Requirement already satisfied: sqlalchemy<3.0.0,>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from langchain_community)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain_community)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied:aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied: multidict>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied: typing_extensions>=4.4 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.8.3->langchain_community)
Collecting langchain-text-splitters<2.0.0,>=1.1.2 (from langchain-classic<2.0.0,>=1.0.7->langchain_community)
  Downloading langchain_text_splitters-1.1.2-py3-none-any.whl.metadata (3.3 kB)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langchain-classic<2.0.0,>=1.0.7->langchain_community)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.0.0->langchain_community)
Requirement already satisfied: langchain-protocol<0.0.14 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.0.0->langchain_community)
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.0.0->langchain_community)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.0.0->langchain_community)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: orjson>=3.9.14 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: websockets>=15.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings<3.0.0,>=2.10.1->langchain_community)
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings<3.0.0,>=2.10.1->langchain_community)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->langchain_community)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->langchain_community)
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->langchain_community)
Requirement already satisfied: certifi>=2023.5.7 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.32.5->langchain_community)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-packages (from sqlalchemy<3.0.0,>=1.4.0->langchain_community)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->langsmith<1.0.0,>=0.1.125->langchain_community)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->langchain_community)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langchain_community)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain_community)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain_community)
Downloading langchain_community-0.4.2-py3-none-any.whl (2.4 MB)
  2.4/2.4 MB 29.0 MB/s eta 0:00:00
Downloading langchain_classic-1.0.8-py3-none-any.whl (1.0 MB)
  1.0/1.0 MB 59.6 MB/s eta 0:00:00
Downloading requests-2.34.2-py3-none-any.whl (73 kB)
  73.1/73.1 kB 5.9 MB/s eta 0:00:00
Downloading langchain_text_splitters-1.1.2-py3-none-any.whl (35 kB)
Installing collected packages: requests, langchain-text-splitters, langchain-classic, langchain_community
```



```
Document(page_content="The solar energy system in modern homes helps balance electricity demand.", metadata={"source": "I1"})
Document(page_content="Python balances readability with power, making it a popular system design language.", metadata={"source": "I2"})
Document(page_content="Photosynthesis enables plants to produce energy by converting sunlight.", metadata={"source": "I3"})
Document(page_content="The 2022 FIFA World Cup was held in Qatar and drew global energy and excitement.", metadata={"source": "I4"})
Document(page_content="Black holes bend spacetime and store immense gravitational energy.", metadata={"source": "I5"}),
]
```

```
from langchain_community.vectorstores import FAISS
from langchain_openai import OpenAIEmbeddings
from langchain_core.documents import Document
from langchain_openai import ChatOpenAI
from langchain.retrievers.multi_query import MultiQueryRetriever
```

```
# Initialize OpenAI embeddings
embedding_model = OpenAIEmbeddings()

# Create FAISS vector store
vectorstore = FAISS.from_documents(documents=all_docs, embedding=embedding_model)

# Create retrievers
similarity_retriever = vectorstore.as_retriever(search_type="similarity", search_kwargs={"k": 5})

multiquery_retriever = MultiQueryRetriever.from_llm(
    retriever=vectorstore.as_retriever(search_kwargs={"k": 5}),
    llm=ChatOpenAI(model="gpt-3.5-turbo")
)
# Query
query = "How to improve energy levels and maintain balance?"

# Retrieve results
similarity_results = similarity_retriever.invoke(query)
multiquery_results= multiquery_retriever.invoke(query)
```

```
for i, doc in enumerate(similarity_results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)

print("\n***150")

for i, doc in enumerate(multiquery_results):
    print(f"\n--- Result {i+1} ---")
    print(doc.page_content)
```



Contextual Compression Retriever

```
from langchain_community.vectorstores import FAISS
from langchain_openai import OpenAIEmbeddings, ChatOpenAI
from langchain.retrievers.contextual_compression import ContextualCompressionRetriever
from langchain.retrievers.document_compressors import LLMChainExtractor
from langchain_core.documents import Document
```

```
# Recreate the document objects from the previous data
docs = [
    Document(page_content=(
        """The Grand Canyon is one of the most visited natural wonders in the world.
        Photosynthesis is the process by which green plants convert sunlight into energy.
        Millions of tourists travel to see it every year. The rocks date back millions of years."""
    ), metadata={"source": "Doc1"}),
    Document(page_content=(
        """In medieval Europe, castles were built primarily for defense.
        The chlorophyll in plant cells captures sunlight during photosynthesis.
        Knights wore armor made of metal. Siege weapons were often used to breach castle walls."""
    ), metadata={"source": "Doc2"}),
    Document(page_content=(
        """Basketball was invented by Dr. James Naismith in the late 19th century.
        It was originally played with a soccer ball and peach baskets. NBA is now a global league."""
    ), metadata={"source": "Doc3"}),
    Document(page_content=(
        """The history of cinema began in the late 1800s. Silent films were the earliest form.
        Thomas Edison was among the pioneers. Photosynthesis does not occur in animal cells.
        Modern filmmaking involves complex CGI and sound design."""
    ), metadata={"source": "Doc4"})
]
```



```
    ), metadata={"source": "Doc4"})  
]
```

```
# Create a FAISS vector store from the documents  
embedding_model = OpenAIEmbeddings()  
vectorstore = FAISS.from_documents(docs, embedding_model)
```

```
base_retriever = vectorstore.as_retriever(search_kwargs={"k": 5})
```

```
# Set up the compressor using an LLM  
llm = ChatOpenAI(model="gpt-3.5-turbo")  
compressor = LLMChainExtractor.from_llm(llm)
```

```
# Create the contextual compression retriever  
compression_retriever = ContextualCompressionRetriever(  
    base_retriever=base_retriever,  
    base_compressor=compressor  
)
```

```
# Query the retriever  
query = "What is photosynthesis?"  
compressed_results = compression_retriever.invoke(query)
```

```
for i, doc in enumerate(compressed_results):  
    print(f"\n--- Result {i+1} ---")  
    print(doc.page_content)
```

